



Contract & SLA Copilot

for Service Operations

An AI copilot that turns 37-minute contract lookups into 6-minute verified answers

NIT / TUHH | Generative AI for Managers | Capstone Pitch | Team NIT26-1 | Ana, Jenny, Ravi

THE PROBLEM

37 minutes to answer a simple SLA question

Service agents spend 37 **minutes on average per inquiry** manually searching through complex, multi-page service agreements and SLA tables.



Slow

37 min per query across 8 manual steps



Error-Prone

Wrong commitments create legal & financial exposure



Inconsistent

Different agents interpret the same clause differently

WHO IS AFFECTED

Service Desk Lead

Customer Service

Daily

Account Manager

Sales / Field

Weekly

Legal Analyst

Legal / Finance

Ad-hoc

~2,600 queries/year | 50 queries/week | 8-step manual process | Growing bottleneck

WHY GenAI — WHY NOW??

From searching documents to answering questions with evidence



Contextual Reasoning

Reasoning Agent evaluates how multiple clauses interact — resolving nuances that rule-based systems miss.



Semantic Intent Matching

RAG matches a user's underlying intent to relevant contract sections — not just keywords.



Dynamic Knowledge

Contract repositories are in constant flux. RAG retrieves from the latest indexed documents — no retraining.



Agentic Escalation

Escalation Agent detects ambiguity or missing data and flags the case for human review automatically.

Traditional keyword search fails because contracts are defined by context, conditions, and exceptions

MVP OVERVIEW

IN SCOPE (4-Week MVP)



F1 Query Structuring

Converts customer question into structured query — contract ID, SLA topic, service type



F2 Clause Retrieval + Citations

Semantic search retrieves relevant contract sections with clause-level references



F3 AI Draft + Human Review

Generates draft response with citations — agents accept, edit, or reject before sending

OUT OF SCOPE



Automatic sending of responses



Multi-language support



Full CRM / ticketing integration



Voice call transcription



Contract editing / lifecycle mgmt



SLA breach prediction



Mobile application

JTBD: "See relevant clauses + a suggested response draft with citations — verify quickly, respond accurately."

ARCHITECTURE — AGENTIC RAG WORKFLOW

Pattern 4: Agentic Workflow — decompose, retrieve, reason, approve, log



KEY COMPONENTS

UI / Copilot Panel

Embedded beside service ticket — displays drafts, citations, Accept/Edit/Reject

Orchestration Layer

Central controller — triggers agents, applies routing, sends flagged cases to human review

Vector Database

Indexed contract/SLA chunks with metadata (version, ID, section) for semantic retrieval

System Logging Layer

Full audit trail — query, clauses, draft, edits, final answer for governance & evaluation

Mandatory HITL gates at Steps 5 & 6 — no AI response reaches a customer without human approval

BEFORE & AFTER — PROCESS TRANSFORMATION

AS-IS | 37 minutes | 8 manual steps

1	Clarify query	4 min	Agent
2	Look up account	5 min	Agent
3	Retrieve contract	6 min	Agent
4	Scan PDFs / SLA tables	8 min	Agent
5	Interpret clauses	6 min	Specialist
6	Draft response	4 min	Agent
7	Review & approve	3 min	Manager
8	Send + log	1 min	Agent



TO-BE | 6 minutes | AI-assisted

1	Submit query	0.5 min	HITL
2	Extraction Agent	0.5 min	AI
3	Retrieval + Reasoning	1.0 min	AI
4	Draft with citations	1.0 min	AI
5	Human review	2.0 min	HITL
6	Specialist (if flagged)	0 min	HITL
7	Send verified answer	0.5 min	HITL
8	Auto-log full trail	0.5 min	AI

Primary bottleneck (Steps 3–5) automated | **84% time reduction** | Full audit trail preserved

VALUE SIZING — ROI MODEL

€133,865

Likely Annual Value

€66,500

Conservative

€133,865

Target

€250,750

Optimistic

Metric	Low	Likely	High
Annual Query Volume	2,000	2,600	3,000
Minutes Saved / Query	31 min	31 min	31 min
Annual Hours Saved	1,033 hrs	1,343 hrs	1,550 hrs
Loaded Hourly Cost	€45	€55	€65
Time Savings Value	€46,500	€73,865	€100,750
Quality / Risk Savings	€20,000	€60,000	€150,000
Total Annual Value	€66,500	€133,865	€250,750

Value per Query: €28.40 | ~50 queries/week × €28.40 = €1,400/week in operational value

30 / 60 / 90 — DAY ROLLOUT PLAN



DAYS 1–30

PILOT

- 10 Tier-1 EMEA agents
- Coverage + SLA + credits only
- 1-day training + prompt library
- Weekly sprint reviews

SUCCESS GATE

≥80% Adoption | Zero Data Leaks | Stable HITL



DAYS 31–60

EXPAND

- Scale to all 50 EMEA agents
- Add Contract Managers (Step 6)
- Fix top 3 pilot issues
- CRM context pre-fills

SUCCESS GATE

≤10 min resolution | ≥85% Accuracy |
Controlled Escalation



DAYS 61–90

SCALE

- SOP — mandatory first step
- Service Ops ownership transfer
- Weekly KPI dashboards
- ROI readout vs €73,865 target

SUCCESS GATE

Formal ROI assessment | ≤6 min AHT | Full audit compliance

Each phase has explicit Go/No-Go gates — disciplined expansion, not premature scaling

KPIs — 11 METRICS ACROSS 4 CATEGORIES

Category	KPI	Baseline	90-Day Target	Type
Adoption	Weekly Active Users	0%	≥85% (~43)	Leading
Adoption	Feature Utilization Rate	0%	≥70%	Leading
Adoption	Avg. Queries/User/Week	0	≥15	Leading
Efficiency	Average Handling Time	37 min	≤6 min	Leading
Efficiency	System Latency	N/A	≤60 sec	Leading
Efficiency	Agent Throughput	1.6 q/hr	≥6 q/hr	Leading
Quality	Human Override Rate	N/A	≤20%	Leading
Quality	Citation Grounding Rate	N/A	≥90%	Leading
Quality	Escalation Accuracy	N/A	≥85%	Leading
Biz Outcome	Operational Value (€)	€0	≥€73,865	Lagging
Biz Outcome	Compliance Audit Pass	N/A	≥95%	Lagging

Leading indicators track adoption & performance early | Lagging indicators confirm realized business value

GOVERNANCE & SAFETY CONTROLS



Data & Access

Confidential classification | SSO/RBAC for 50 agents |
100% sessions tied to user ID



Logging & Audit

Full trail: query, clauses, AI draft, agent edits, final
response, latency



HITL Approvals

Gate 1: Agent reviews citations | Gate 2: Contract
Manager for flagged ambiguities



Evaluation Set

100-case Golden Set — 40% coverage, 30% SLA, 30%
credits — including edge cases

STOP-AND-FIX TRIGGERS

Answer Accuracy	< 85%	Pause + review failures
-----------------	-------	-------------------------

Citation Grounding	< 90%	Fix retrieval/chunking
--------------------	-------	------------------------

Override Rate	> 25%	Refine prompts
---------------	-------	----------------

Escalation Rate	> 30%	Improve coverage
-----------------	-------	------------------

System Latency	> 90 sec	Optimize infra
----------------	----------	----------------

Data Security	Any HIGH leak	Immediate suspend
---------------	---------------	-------------------

Compliance	< 95% audit	Legal review
------------	-------------	--------------

Kill-Switch: Product Owner + Service Ops Lead

No AI response reaches a customer without explicit human review and approval

THE ASK

MVP Investment

€38K – €55K

Infrastructure: €8–12K

Development (4 wks): €25–35K

Pilot monitoring: €5–8K

Annual Return

€133,865

Payback within Year 1

Time savings: €73,865/yr

Risk avoidance: €60,000/yr

WHAT WE NEED TO START

1. Legal sign-off on disclaimer language and escalation paths
2. Read access to contract repository (SharePoint / DMS)
3. 10 pilot agents identified from EMEA Tier-1 team
4. 4-week sprint approval with small cross-functional team

This is not an AI experiment — this is a controlled operational upgrade with measurable ROI



Contract & SLA Copilot

37 min → 6 min | €133,865 annual value | Payback in Year 1

Thank you...

Pitch Video: <https://drive.google.com/file/d/1xaQLj9b4IKRLHEA-RCLk2UWHIKsDk4zA/view?usp=sharing>

Mini MVP: <https://contractcopilot.streamlit.app/>

Github Repo: https://github.com/ravabulls/mini_mvp